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1 January 1971

SUPERSEDING
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17 February 1955

MILITARY SPECIFICATION

MANUALS, TECHNICAL: INTERMEDIATE MAINTENANCE INSTRUCTIONS

(For Avionics Equipment and Systems)

1. SCOPE

1.1. This specification prescribes the requirements for the preparation of technical manuals covering intermediate maintenance of an item of equipment contained in, or a part of, (a) automatic flight controls; (b) communications systems; (c) guidance and navigation systems; (d) bombing—navigation systems; (e) fire control systems; (f) electronics counter measure systems; (g) photographic and reconnaissance systems; and (h) nonnuclear weapons and releases.

2. APPLICABLE DOCUMENTS

2.1. *Government documents.* The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-38784	Manuals, Technical: General Requirements for Preparation of
MIL-P-38790	Printing Production of Technical Manuals; General Requirements For
MIL-M-38799	Manuals, Technical: Block Diagrams and Maintenance Dependency Charts
MIL-STD-836	Lubrication of Military Equipment

HANDBOOK

MIL-HDBK-275	Guide for the Use of Lubricants, Compounds, and Fluids in Aircraft

PUBLICATIONS

Technical Order

00-25-203	Standards and Guidelines for the Design and Operation of Clean Rooms and Clean Work Stations
00-25-234	General Shop Practice Requirements for the Repair and Test of Electronic Equipment.

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1. *Preparation.* The general manner of preparation shall conform to MIL-M-38784 and MIL-P-38790. Every effort shall be made to include only unclassified information in the manual. The manual shall be prepared in the form (manuscript, reproducible copy, negatives) specified in the contract.

3.2. *Contents.* The instructions shall provide all essential information required by personnel for performing intermediate maintenance on the components of the equipment. The instructions shall show the interface between the components and the end item equipment. Maintenance instructions shall enable personnel to test, troubleshoot, repair, and return it to an acceptable performance standard, all components of the equipment in a minimum of time with the tools, test equipment, and spare parts authorized for intermediate maintenance activities. Instructions shall cover all items comprising the equipment, such as accessories, components, wiring, junction boxes, etc. Procedures requiring a team of specialists for the accomplishment of maintenance tasks shall have the tasks of these individuals designated separately in sequence of operation.

3.2.1. *Development of text.* The requirements of MIL-M-38784 are applicable. Repetitious information shall be avoided except for clarity or where considerable maintenance time would be saved by repeating information to preclude excessive cross referencing between manuals. Emphasis shall be placed on the specific steps of the maintenance procedures to be followed, the results which may be expected or desired, and the corrective measures required when such results are not obtained.

3.2.2. *Environmental control.* Environmental control information shall be provided. The manual shall include information on any special maintenance required under extreme temperature, altitude, and humidity conditions within the limits established by the design specification for the equipment. Refer to TO 00-25-203.

3.2.3. *Nomenclature.* Nomenclature shall be in accordance with the requirements of MIL-M-38784.

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3.2.4. *Reference to parts of the equipment.* Reference to parts of the equipment, and its components, etc, shall be by type designation, standard nomenclature, reference designation, or figure and index number, as applicable.

3.2.5. *Abbreviations.* Abbreviations shall be in accordance with MIL-M-38784.

3.2.6. *Illustrations.* Illustrations and diagrams shall be provided in accordance with MIL-M-38784.

3.2.6.1. *Illustration keys.* Legends, tables or text applicable to partial page illustrations shall, whenever possible, appear on the same page. For a full page illustration, the information shall appear on a facing page or the illustration, legend, table, and text may be arranged as a foldout page. If the legend, table, or text exceeds the allowable space, it may run over to a second page.

3.2.6.2. *Reference designations.* Electrical and electronic reference designations shall be in accordance with MIL-M-38784.

3.2.6.3. *Diagrams.* The manual shall include diagrams and associated data necessary for intermediate maintenance personnel to: (a) understand the make-up and function of each intercomponent, power, control, signal circuit, and of each mechanical system; (b) trace each circuit and mechanical system; (c) accomplish general and specific troubleshooting on inoperative or malfunctioning circuits or mechanical systems. Foldout diagrams shall be in accordance with MIL-M-38784.

3.2.6.3.1. *Explanatory text.* An explanation of the diagrams used in the manual shall be given as to their content and proper use. This shall include the working relationship to material in other sections, how to locate a particular diagram, an explanation of the wiring identification system used on the diagrams, and any other information considered essential for proper understanding and use.

3.2.6.3.2. *Symbols.* The requirements of MIL-M-38784 are applicable.

3.2.6.3.3. *Schematic diagram general requirements.* Where the equipment's nature, relative complexity, and number of components make it practicable, a single schematic diagram shall be prepared to cover all signal circuits, power circuits, and control circuits. When a single diagram is impracticable, a separate diagram shall be prepared for each item of equipment. Any circuits left in "mid-air" and picked up on other diagrams and which terminate with a plug, terminal strip, jack, etc, shall show a composite plug grouping, and each lead shall be extended through the plug, terminal strip, jack, etc, and labeled to indicate the necessary designator of the first connection within the other component, or other significant point of termination.

3.2.6.3.3.1. When the equipment obtains information from, or provides information to other items of equipment of a system, diagrams shall be provided showing all required interconnections.

3.2.6.3.3.2. Power circuits, control circuits and signal flow within a particular circuit(s) or functional loop(s) shall be identified. Any use of color shall be in accordance with MIL-M-38784.

3.2.6.3.4. *Schematic diagrams for equipment.* A schematic diagram of an item of equipment shall show the following:

a. Subassemblies shall be indicated by a series of broken lines.

b. For each detail part, identification shall be by reference designations.

c. For each detail part, the principal electrical characteristics expressed in actual values (ohms, microfarads, microhenrys, etc.), shall be tabulated next to the diagram and indexed by reference designation to applicable points on the diagram when sufficient space is not available on the diagram itself. Values normally shall be expressed as numbers, omitting the unit of measurement (such as ohms). However, a general note shall state which units are used.

d. Wave shapes, duration, and amplitude of voltages at designated test points throughout the circuit and at interconnecting points within the equipment or systems, under normal test conditions and other specified conditions shall be indicated on the schematic diagram of the component.

3.2.6.3.5. *Overall schematic diagrams.* If separate component diagrams are not provided, the overall equipment schematic shall show the information called for in 3.2.6.3.4., and in addition thereto, interequipment circuits showing the connections and terminals properly numbered, with a series of broken lines indicating items of equipment.

3.2.6.3.6. *Power and control schematics.* Separate schematic diagrams shall be prepared, as follows, to show the power and control circuits when it is impracticable to show such information on the overall schematic diagram: each power circuit and each control circuit with its relationship to the signal circuits using a combination of blocks, schematics, or pictorial diagrams; tolerances of power distribution associated with each power and control circuit.

3.2.6.3.6.A. *Logic diagrams.* When appropriate to the design of the equipment, overall operation of the equipment shall be shown by means of conventional logic diagrams. Current generation of equipment utilizing solid state and integrated circuit devices is understood and troubleshooting is accomplished by means of logic diagrams rather than detailed schematics.

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3.2.6.3.7. *Waveform diagrams.* If not provided on functional and schematic diagrams and required in support of maintenance tasks, these diagrams shall show the waveforms and nominal values at designated points (Actual photographs, where possible) for normally functioning equipments, or systems as seen in an oscilloscope of a type listed in Section II of the manual. Operating conditions and switch positions of the operating equipment shall be noted at the time of each indication.

3.2.6.3.8. *Interconnection cable diagrams.* These diagrams shall show the cabling or connections for the system, equipment, or component. Interconnections between components or equipments shall be indicated by either a diagram or a table.

3.2.6.3.9. *Combination diagrams.* Combinations of the information required by 3.2.6.3.7. and 3.2.6.3.8. may be shown on a single diagram, where practicable.

3.2.6.3.10. *Module schematic diagrams.* Schematic diagrams of electronics modules, when shown separately, shall be arranged to indicate signal flow progressing from left to right. Where possible, an equivalent block/logic diagram of the module circuitry shall be placed on the schematic to aid in interpreting the schematic. (See figure 5)

3.2.6.5. *Test point identification.* Test point identification shall be incorporated on schematic wiring diagrams, tables, charts and illustrations. Test point identification will not be required for equipment with self-test capability, equipment designed to be tested by an automatic checkout system, and equipment designed with critical test points accessible at front panel connectors. Where the design of the equipment includes built in test capability, tests performed using this capability will be identified but not outlined in detail. Any function requiring the participation of maintenance personnel, or interpretation of test results, shall be included in detail. Instructions required to determine the operability of the design built in test circuits and components, fault isolation and repair instructions, shall be included. Symbols shall be as marked on the equipment, or as they appear on applicable engineering drawings. Voltages, waveforms, etc, should be placed adjacent to the test point and if required may be in tabular form. Indiscriminate use of test points shall be avoided; for example, test point identification shall normally be restricted to signal tracing test points and voltage and continuity check points. The following test point identification system is commonly used.

3.2.6.5.1. *Major test points.* A star-enclosed Arabic numeral, ☆ is used to identify and designate the test points used in checking the overall functions of, and localizing trouble to individual components, or groups of components. Major test points are identified on schematic wiring diagrams, tables, charts, and illustrations by use of the Arabic numerals, 1, 2, 3, etc, enclosed in a star; but will be referred to in the text as test point 1, test point 2, etc. Typical examples of major test points are: (a) the input terminals for the supply source voltage; (b) the high voltage output terminals of the equipment's power system; (c) the signal input terminals of the equipment; (d) the signal output terminals of the equipment; and (e) the major points of power or voltage distribution within the equipment.

3.2.6.5.2. *Secondary test points.* An encircled capital letter, Ⓐ, is used to identify and designate the test points used in isolating causes of subnormal performance within a specific assembly or subassembly of the equipment. Secondary test points are identified on schematic wiring diagrams, tables, charts, and illustrations by use of capital letters, A, B, C through Z; AA, AB through AZ; BA, BB through BZ, etc, enclosed in a circle; but will be referred to in the text as test point A, test point B, etc. Typical examples of secondary test points are: (a) circuit supply voltage terminals; (b) signal output points used in checking the gain of a stage or group of stages.

3.2.6.5.3. *Minor test points.* An encircled capital letter and Arabic numeral, Ⓐ1, is used to identify and designate the test points used in isolating causes of abnormal indications within a specific circuit of the equipment. Minor test points are identified on schematic wiring diagrams, tables, charts and illustrations by use of a capital letter and Arabic numeral, A1, A2, etc, enclosed in a circle but will be referred to in the text as test point A1, test point A2, etc. Typical examples of minor test points to be so designated are: (a) specific voltage and resistance check points; (b) all other points used in checking for trouble in a specific circuit. Different letters are assigned to each component.

3.2.6.5.4. *Assignment of symbols.* Test point identification symbols on the schematic wiring diagrams shall be assigned consecutively throughout the composite equipment. Once assigned, a symbol shall not be reassigned to another test point. An explanation of the use of test points shall appear in the Introduction of the manual.

3.2.7. *Cable fabrication.* When cable assemblies are not supplied, but are required for bench test set-up, instructions shall be provided for fabricating interconnecting cable assemblies from spares and bulk supplies. The part number, drawing number, and manufacturer or designer for each part of the cable assembly shall be shown, as well as identification of wires, connectors, pin connections, and letters or other designators. .

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3.3. *Arrangement.* The manual shall be arranged as follows: Each section shall begin on a right hand page. When a section is not applicable, the section title shall appear where the section would normally begin, with a notation, "Not Applicable". The next section shall start on the same page. Where a section does not apply, the table of contents for the manual shall so indicate with the statement "Section_____, Not Applicable." Also refer to 3.3.6.4.

	Front Matter
Section I	Description
Section II	Special Tools and Test Equipment
Section III	Theory of Operation
Section IV	Description of System Tie-in of Equipment and accessories
Section V	Checkout and Troubleshooting
Section VI, VII, VIII, etc	Maintenance Instructions (one section for each unit component)
Last Section	Alphabetic Index

3.3.1. *Front matter.* Front matter shall be prepared in accordance with MIL-M-38784. When special techniques or materials are required, the introduction shall include a reference to TO 00-25-234 for instructions concerning procedures and specific materials to be utilized in soldering such electronic equipment.

3.3.2. *Section I, Description.* This section shall contain the following information.

3.3.2.1. A composite illustration of the separately housed components, giving their nomenclature, and if applicable, Government type designations.

3.3.2.2. Major characteristics of the equipment or components.

3.3.2.3. A brief account of the equipment's purpose, limitations, and how, in general, it accomplishes its mission. Other general information essential for intermediate maintenance instructions.

3.3.2.4. Condensed factual data (tabular form where possible) to familiarize maintenance personnel with the equipment's characteristics and physical make-up. The following items are typical points to be covered, as applicable.

a. Tube complement and solid state devices by component: Quantity, Government type number, function.

b. Fuse complement by component: Quantity, type, number, ampere rating.

c. Frequency characteristics, number of preset frequencies, frequency stability.

d. Output characteristics: (voice, pulse, continuous wave(cw), modulated continuous wave (mcw) etc.)

e. Tuning band ranges, sensitivity, and selectivity.

f. Power requirements: Voltage, type, frequency and tolerance.

g. Current required at specified rated voltage for: (1) filament; (2) dynamotor; (3) inverter; (4) motor; (5) other components; (6) normal operation total; (7) standby; (8) starting maximum.

h. Power-generating equipment (rotary, vibrator, battery, converter) date, manufacturer's name, type, output voltage, current rating, permissible ambient temperature, and temperature rise.

i. Location and function of operating and adjustment controls used in normal operation.

j. Length of stroke, operating pressure, displacement, duty cycle, inlet and outlet connection sizes, direction of rotation, etc.

3.3.3. *Section II, Special Tools and Test Equipment.* This section shall list and illustrate the special tools and test equipment required for the work described. Such items shall be selected from the list of tools and test equipment approved by the Government by means of AGE Listings, Provisioning Conference, Engineering Change Proposals (ECPs), Procurement Documents, etc. Standard types of tools, such as screwdrivers, pliers, shall not be listed. Standard types of test equipment, such as voltmeters, tube testers, shall be listed but not illustrated.

3.3.3.1. *Special tools list.* The list of special tools shall be arranged in the following format. Tools not illustrated in use in other sections of the manual shall be illustrated following the Special Tools List in this section.

SPECIAL TOOLS LIST

Part (Tool) Number	*Figure & Index No.*	*Nomenclature*	*Use*

a. *Part (tool) number.* Tools shall be listed in alpha-numeric sequence by part (tool) number, manufacturer's five digit code, or manufacturer's name and city.

b. *Figure and index number.* The figure and index numbers that appear in other sections, or this section, of the manual, shall be shown.

c. *Nomenclature.* Refer to 3.2.3.

d. *Use.* The purpose of each tool shall be stated. If the use is described in any part of the manual, reference to the applicable paragraph will suffice.

3.3.3.2. *Test equipment list.* The list of test equipment shall be arranged in the following format. Test equipment not illustrated in use in other sections of the manual shall be illustrated following the Test Equipment List in this section (standard test equipment excepted).

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TEST EQUIPMENT LIST

Type Designation	*Alternate Type Designation*	*Figure & Index No.*	*Nomenclature*	*Use*

a. *Type designation.* Test equipment shall be listed in alpha-numeric sequence by type designation, using the AN type designation, if assigned, manufacturer's five digit code, or manufacturer's name and city.

b. *Alternate type designation.* If an alternate item of test equipment can be used, its AN type designation, if assigned, or the commercial or manufacturer's designation, shall be listed.

c. *Figure and index number.* The figure and index numbers that appear in other sections, or this section, of the manual, shall be shown.

d. *Nomenclature.* Refer to 3.2.3.

e. *Use.* The purpose of each item of test equipment shall be stated. If the use is described in any part of the manual, reference to the applicable paragraph will suffice.

3.3.4. *Section III, Theory of Operation.* See figures 1, 2, and 3. This section shall concisely explain theory of operation of the equipment and accessories. The explanation shall be accomplished by means of functional diagrams accompanied by explanatory text. A series of diagrams, from one that shows gross overall operation down to those showing detail functions, may be supplied (in pyramid style) for complex equipment such as shown in the block level illustration of Figure 1. A master functional diagram shall be provided which portrays on a single page, the overall function of the equipment and accessories. The purpose of this diagram shall be to describe the operation of each major functional division in relationship to other functional divisions such that the total operation is understood, and access to the next subordinate level of functional diagrams is provided.

3.3.4.1. When the functional divisions defined on the master block diagram are sufficiently complex, further division is necessary. In this case, intermediate and functional diagrams shall be provided to show the transition between major divisions on the master functional diagram and the subordinate functional diagrams necessary to describe the theory of operation. Intermediate functional diagrams shall be prepared as necessary by dividing each functional block on the master functional diagram into discreet single page subfunctional diagrams.

3.3.4.2. As necessitated by equipment complexity, block diagrams shall be prepared to further show the subfunctions of the equipment.

3.3.4.3. The composite of the functional diagrams shall be constructed to provide the user the clearest explanation of theory and understanding of the maintenance requirements of the equipment and accessories.

3.3.4.4. Basic theory of operation, shall be excluded; however, special or unique features shall be explained and illustrated.

3.3.5. *Section IV, Description of System Tie-In of Equipment and Accessories.* This section shall explain the integration of all equipments and accessories into an operational system. When a system uses inputs (signals/voltages) from a different system or provides outputs (signals/voltages) to a different system, input sources and output destinations shall be explained and shall be shown in block diagrams. The explanation shall be a narrative description of the tie-in of the equipments and accessories of the system.

3.3.6. *Section V, Checkout and Troubleshooting (Also refer to 3.3.6.4.)* The manual shall contain instructions for performance testing and for locating and identifying the cause of malfunction. Wherever possible this shall be done on a unit basis, in which case the instructions shall appear in the applicable unit maintenance sections of the manual. However, where it is necessary to combine two or more items of equipment into a system, or portion thereof for performance testing or to locate and identify cause(s) of malfunction the instructions shall be placed in this section of the manual. In either case, the instructions shall be as follows:

3.3.6.1. *Bench test set-up (when applicable).* Figures shall be presented illustrating the suggested bench test set-up showing: (a) the details of intercomponent cabling and (b) cabling between test equipment and equipment or component under test for each procedure. Where a number of separate procedures utilize identical test set-ups, one figure illustrating the set-up shall be presented once, and not repeated. Appropriate reference to the figure number will suffice for each procedure. If a cable is to be connected to more than one connector during the procedure, an asterisk or other symbol should be placed at the unconnected connector(s) on the test setup diagram to indicate that connection instructions will be given in the appropriate step of the test procedure. Whenever such materials is too voluminous to be practical for inclusion in the manual, reference will be made to the applicable technical order as noted in MIL-M-38784.

3.3.6.2. *Analysis.* A malfunction isolation guide shall be provided, in a logical step-by-step troubleshooting sequence which relates AGE and/or component indications to specific detail parts or circuits which are most likely to have caused the problem.

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3.3.6.3. *Test Equipment.* Instructions essential for proper use of test equipment in diagnosing trouble within a specific item of equipment shall be included within each procedure.

3.3.6.4. *Alternate method of presenting troubleshooting data.* When the procuring activity so indicates in the contract, troubleshooting data shall be prepared separately in accordance with the requirements of MIL-M-38799. When such data are separately prepared, Section V, Troubleshooting, shall consist of one page and a reference to the publication number of the manual that contains the troubleshooting information.

3.3.7. *Section(s) VI, VII, VIII, IX, etc. Maintenance Instructions.* This section shall describe and illustrate any tasks necessary (installation of equipment into jigs, special fixtures, etc.) to prepare equipment for maintenance. The removal of replacement spares from containers shall be described, as applicable. Intermediate maintenance includes pre-operational checks (safety, continuity, etc), malfunction identification chart, malfunction isolation procedures, schematic diagrams, wiring diagrams, and circuit analysis, which are used to isolate a malfunction to an assembly, subassembly, or detail part. These sections shall also contain information necessary to disassemble, clean and inspect, repair, lubricate, reassemble, align and test the equipment.

3.3.7.1. *Limited maintenance instructions.* Information shall be included to provide limited repair instructions thereby eliminating the need for complete testing and alignment of the component. The assemblies and parts which can be placed in limited maintenance shall be listed in a table, the checkout and alignment procedures required after repair shall be stated.

3.3.7.2. *Inspection.* Inspection instructions shall be provided to prolong service life and assure continued operation. Specific characteristics and points to be inspected shall be included. Inspection of disassembled equipment for damage/wear shall be described. This information shall be tabular or chart form, wherever applicable, with emphasis on allowable service limits, wear, backlash, end play, balance, length and depth of scoring, etc. These tolerances are not to be confused with manufacturing tolerances; they are tolerances that may or may not exceed manufacturing tolerances, yet will not impair performance. This information shall be sufficiently comprehensive and adequate to serve as standards by which intermediate maintenance technicians may determine when parts may be continued in use and when they must be replaced.

3.3.7.3. *Malfunction isolation procedures.* These procedures shall be provided as follows:

3.3.7.3.1. *Prefunctional checks.* Data is required to provide checks in "Power Off" condition. The information shall be tabulated in three columns:

Circuit Being Checked	*Value*	*Part Being Checked*

The following note shall precede the table: "NOTE: It is not necessary to perform the following prefunctional checks unless reported deficiency of equipment to be repaired indicates the internal shorted or open circuit exists." Procedures for manually testing electromechanical equipment shall also be provided when applicable.

3.3.7.3.2. *Functional checks.* A listing of equipment switch controls, and test equipment switch and control settings shall be provided. Functional checks shall be in the following tabular format unless the procuring activity requests otherwise. Procedural steps which have no normal indication requirement may extend across all columns.

Step	*Procedure*	*Normal Indication*	*Corrective Action*

3.3.7.4. *Disassembly.* Detailed instructions shall be provided for separating or disconnecting assemblies unless the method is obvious.

3.3.7.5. *Detail parts and assembly checkout.* Detailed instructions shall be provided for visual inspection and use criteria and for electrical inspection, if applicable. Data shall be in tabular format, if practical. Connector pin location diagrams shall be provided when pin numbers cannot otherwise be located. See Figure 4.

3.3.7.6 *Cleaning.* Special cleaning methods shall be described. Cleaning agents shall be identified. Instructions and specifications shall be given for any cleaning methods to be used. Use of corrosion preventive compounds on all steel parts after cleaning shall be stated.

3.3.7.7. *Lubrication.* Data for lubricating disassembled equipment shall include the specifications and data for grease, oils, etc. Use of special tools for lubrication shall be illustrated. Refer to MIL-HDBK-275 and MIL-STD-838.

3.3.7.8. *Reassembly.* Detailed instructions for reassembly shall be provided, together with the necessary inspections as reassembly is accomplished, and the final inspection instructions upon completion of reassembly.

3.3.7.9. *Alignment checkout.* Detailed step by step instructions for alignment checkout shall be included. Tabular data shall be prepared in the following format. Procedural steps which have no normal indication requirement may extend across all columns.

Step	*Procedure*	*Normal Indication*	*Corrective Action*

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3.3.7.10. *Circuit analysis (detailed).* Data shall provide input, internal, and output signal flow and the functions of assemblies, subassemblies, and detailed parts of the equipment in normal operating condition. Signal and voltage levels may be given to clarify operation. Schematic diagrams of partial circuits or stages shall be prepared for clarity.

3.3.7.11. *Final test.* Instructions shall be given for conducting tests of the reassembled equipment.

3.3.8. *Alphabetical index.* When the manual exceeds 50 pages, an alphabetical index shall be provided. The requirements of MIL-M-38784 are applicable.

4. QUALITY ASSURANCE PROVISIONS

4.1. Quality assurance provisions shall be in accordance with the requirements of MIL-M-38784 and MIL-P-38790.

5. PREPARATION FOR DELIVERY

5.1. Packaging, packing, and marking for shipment shall be in accordance with the requirements of MIL-M-38784 and MIL-P-38790.

6. NOTES

6.1. *Intended use.* The manuals prepared in accordance with this specification are intended for use by personnel in the intermediate maintenance of avionics equipment and systems.

6.2. *Ordering data.* Procurement documents should specify:

a. Title, number, and date of this specification.

b. Form in which manuals shall be provided (3.1)

c. Whether functional checks shall not be in tabular format (3.3.7.3.2.).

d. Alternate method of presenting troubleshooting data (3.3.6.4.).

Custodian:
Air Force—16

Preparing Activity:
Air Force—16

Review Activities:
Air Force—01, 10, 11, 13, 70, 71, 80, 82, 84

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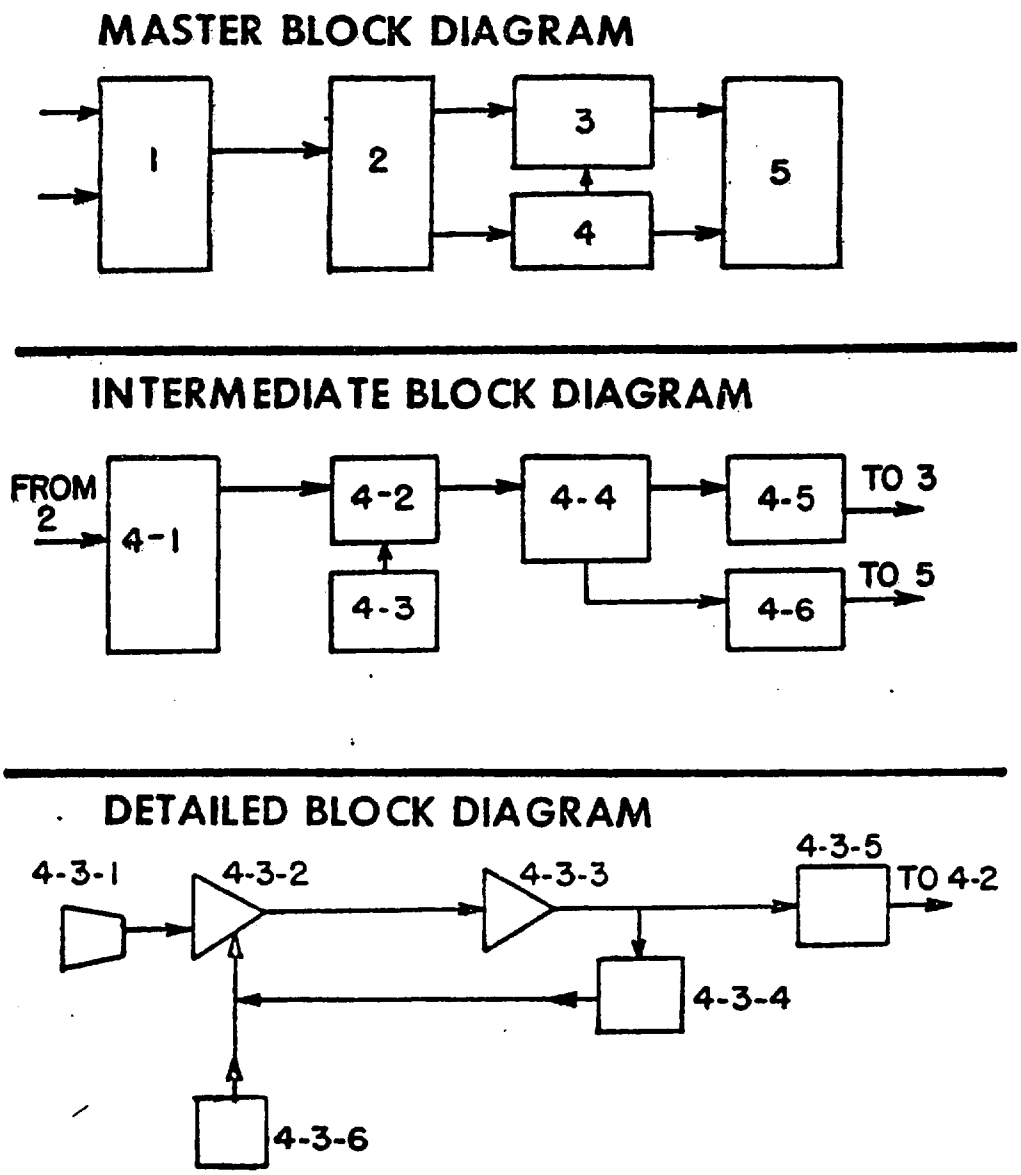


Figure 1. Functional Diagram Pyramid.

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Coincidence Amplifier.

The coincidence amplifier detects the drift of the failure-warning test problem solution in either direction for both PLAN and PROFILE modes of operation. As a result it maintains control of the displayed failure-warning cursor (minimum range mark in PLAN mode, horizontal reference line in PROFILE mode). When the test problem solution remains within set-in upper and lower tolerance limits, the failure-warning cursor will be displayed on the Terrain Display Indicators. If the tolerance limits are exceeded, the failure-warning cursor will be removed from the display of the Terrain Display Indicators and the TA WARNING indicators will light up. Provisions are made within the coincidence amplifier for adjusting the PLAN and PROFILE tolerance limits individually and for automatic switching of these limits as a function of the mode selected. The coincidence amplifier also samples the $+E_S$ gate for both modes of operation and requires that the gate be present and that the gate potential be maintained above a specific level for proper gating of other Terrain Computer modules.

The ultimate function of the coincidence amplifier is to keep the failure-warning relay K6013 energized when the solution of the failure-warning test problem is within the upper and lower tolerance limits, and the problem solution (clamped $\triangle H$ in PROFILE mode, or $\triangle HE_S$ in PLAN mode) is time coincident with the $+E_S$ gate and the failure-warning gate. When energized, relay K6013 completes the circuit for the minimum range mark to the gated video amplifier so the minimum range mark will be displayed in the PLAN mode of operation and it provides a ground path for the Unblanking Pulse Generator which provides the signal for the horizontal reference mark in the PROFILE mode of operation. If the $\triangle HE_S$ or clamped $\triangle H$ is not within tolerance, or if the $+E_S$ gate or failure-warning gate is not present, the coincidence amplifier will not have an output and the failure-warning relay K6013 will deenergize, removing the minimum range mark or the horizontal reference line from the display on the Terrain Display Indicators. When failure-warning relay K6013 is deenergized, it provides a ground path for the TA WARNING indicators, causing them to light.

The functions of the coincidence amplifier are checked and necessary adjustments are performed in the Plan and Profile Coincidence Amplifier Limits Adjustments procedure in section 27 of T O 11B31-2-6-2-2 using the AN/ASM-46B Computer Test Set and the 545A oscilloscope.

Figure 2. Theory of Operation Example (Sheet 1 of 3)

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AND Gate CR55 and CR56. The coincidence amplifier performs it function in the following manner (see adjacent figure). The +Eg gate is applied to the AND gate circuit, consisting of diodes CR55 and CR56, through a nonlinear delay circuit and a direct-coupled amplifier. The nonlinear delay circuit is so biased that the +Eg gate must exceed 20-volts amplitude before it will turn on the direct-coupled amplifier. The nonlinear delay circuit delays the +Eg gate sufficiently to cause sampling of the ΔHE_g pulse to be delayed until leading edge spiking has decayed to an insignificant value. The failure-warning gate is applied to the other input of the AND gate circuit. When the +Eg gate and the failure-warning gate are time coincident and the +Eg gate has sufficient amplitude, the output of the AND gate will be a positive pulse. The positive pulse is applied to a buffer stage and the output of the buffer stage is applied to AND gate diodes CR59 and CR60.

ΔH and ΔHE_g Signal Selection. Relay K51 in the coincidence amplifier selects the HE_g pulse in the PLAN mode of operation or the clamped ΔH in the PROFILE mode of operation. The clamped H signal will be considered. The clamped ΔH is applied to a two stage buffer consisting of transistors Q53 and Q54. The output of the buffer is a -1600-millivolt pulse when the clamped ΔH is -1600 millivolts in amplitude. This -1600-millivolt pulse is applied to two AND gate diodes, CR57 and CR58. The output of one AND gate (CR57 and CR59) is applied to the lower limit detector Q57, the output of the other AND gate (CR58 and CR60) is applied to the upper limit detector Q56.

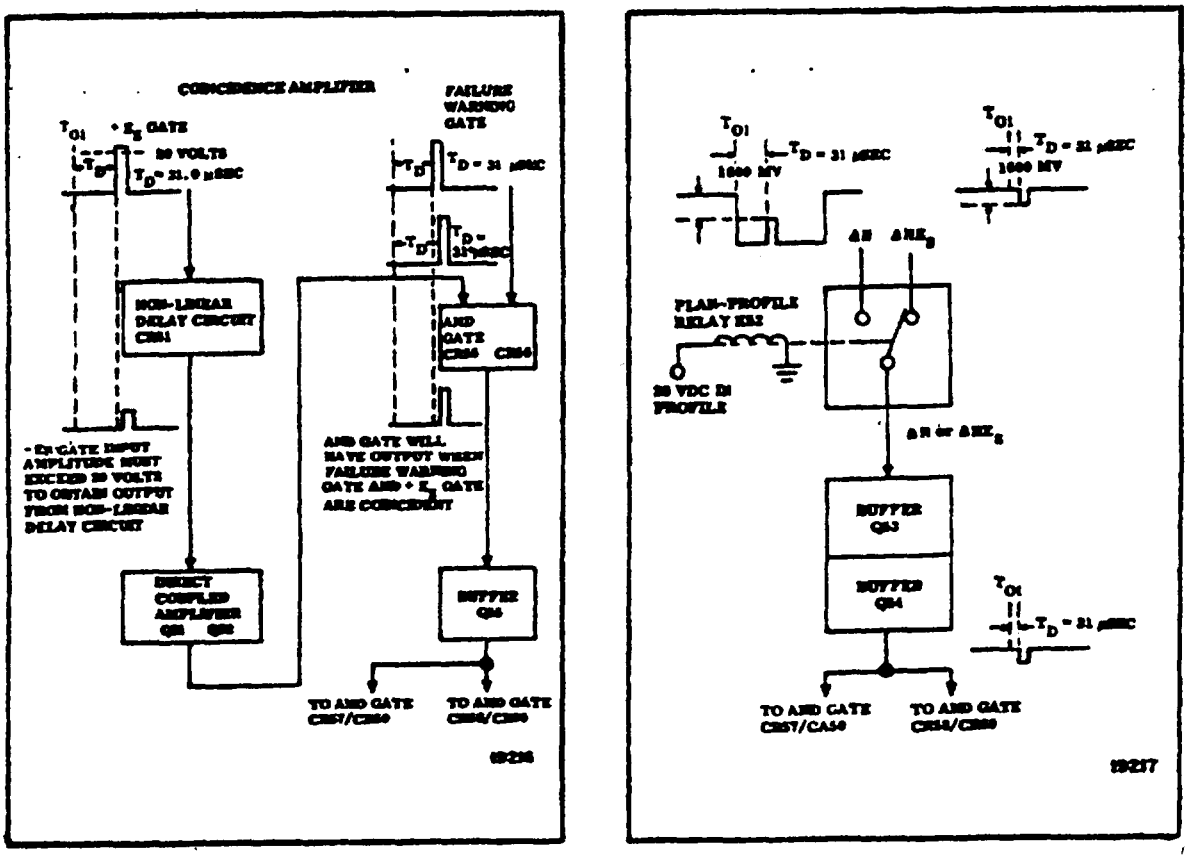


Figure 2. Theory of Operation Example (Sheet 2 of 3)

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Quiescent Voltage Levels Between Pulses. (See figure 4-352N.) The emitter of the lower limit detector Q57 is biased to -1900 millivolts, and the emitter of the upper limit detector Q56 is biased to -1300 millivolts. The potential applied to the bases of both detectors, between input pulses from AND gates, is such that both detectors conduct heavily. The collector potential at both detectors is near zero. For the lower limit detector Q57, this low potential is applied directly to the base of AND gate Q59, holding it near cutoff. For upper limit detector Q56, this low potential is applied to the base of inverter Q58, holding it near cutoff. The potential at the collector of Q58 is high. This high potential is applied to the base of AND gate Q60, causing it to conduct heavily. The potential at the collector of AND gate Q60 is near zero. This low potential is applied to the input of the multivibrator, holding it at cutoff.

In the plan and profile coincidence amplifier limits adjustments procedure, potentiometer R78 is adjusted to provide the -1300 millivolt upper limit bias applied to the base of transistor Q56 in Plan mode; potentiometer R79 is adjusted to provide the upper limit bias during Profile mode. Potentiometer R86 is adjusted to provide the -1900 millivolt lower limit bias applied to the base of transistor Q57 in Plan mode; potentiometer R87 is adjusted to provide the lower limit bias during Profile mode. Relay K52 provides the switching function to connect potentiometers R78 and R86 into the circuit during Plan mode, and to connect potentiometers R79 and R87 into the circuit during Profile mode.

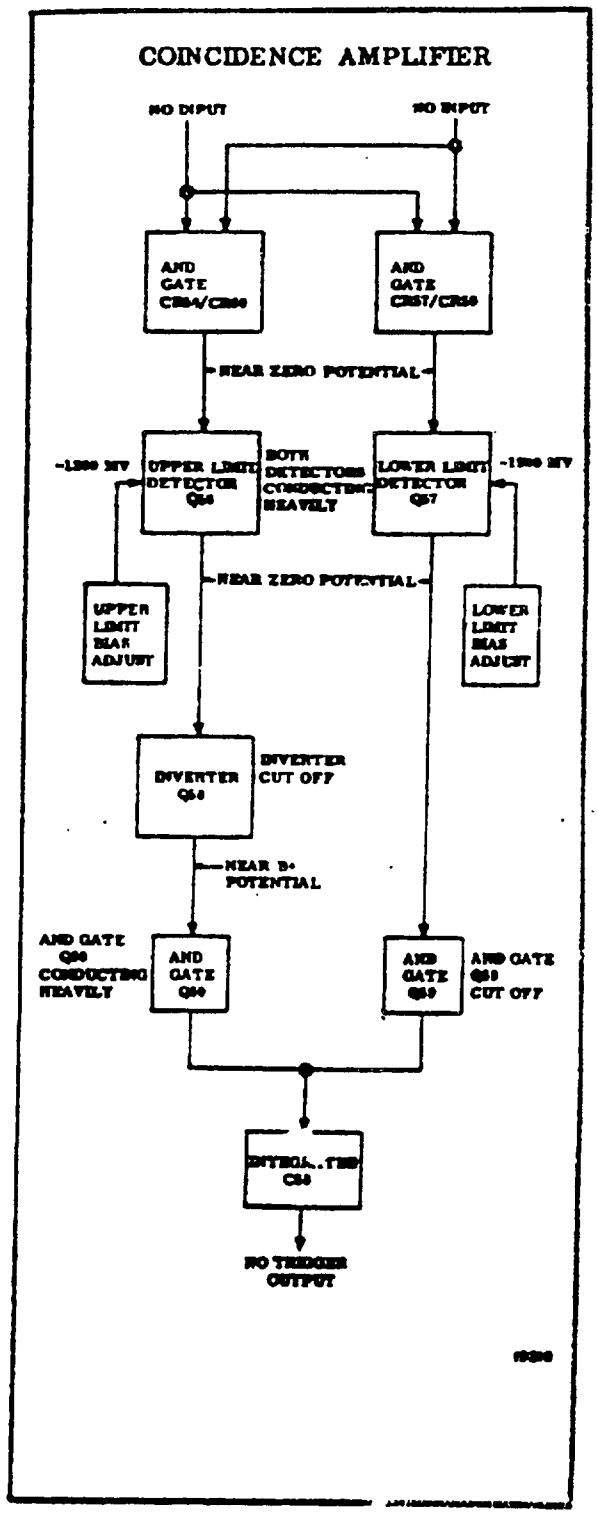
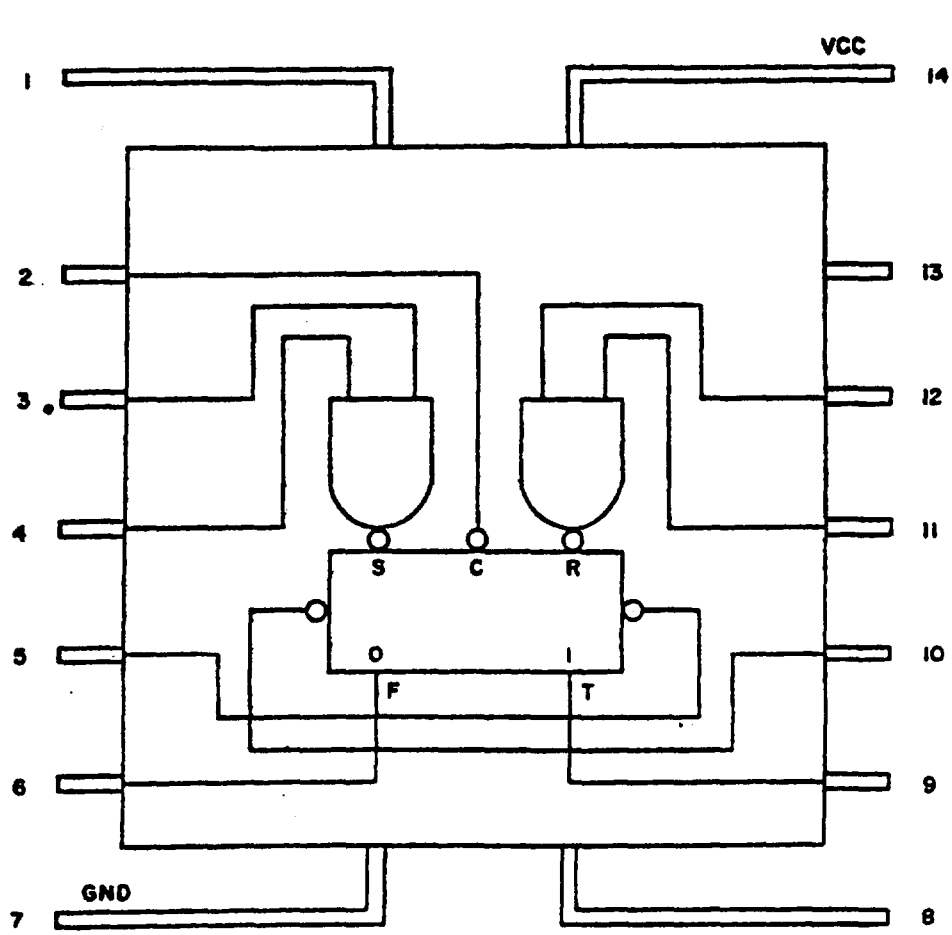


Figure 2. Theory of Operation Example (Sheet 3 of 3).

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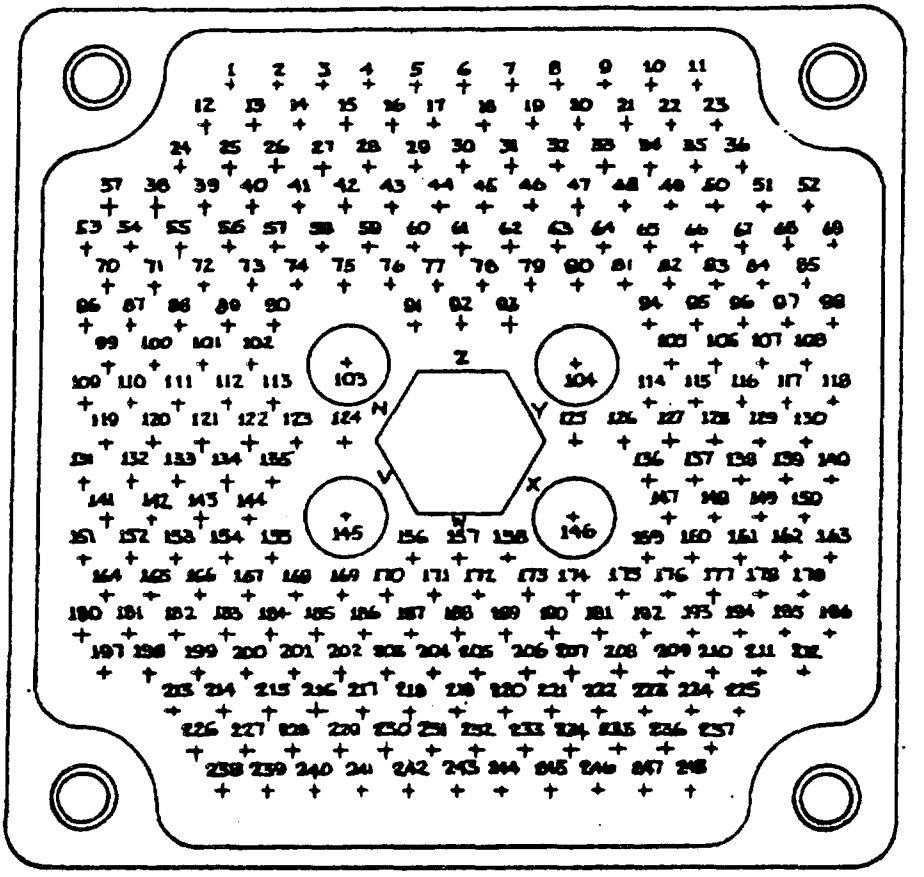


CLOCK GATED FLIP FLOP
251129

4-35. The clock gated flip-flop consists of two NAND gates cross coupled so that the output of each is fed back to the input of the other as shown in figure 4-1. In addition to the two NAND gates other circuitry is present to provide an input for a clock pulse which inhibits a change of state unless a clock pulse is applied. A means for quick setting and resetting (pins 10 and 5 respectively) is also provided. Pins 7 and 14 are used to supply bias voltage to the flip-flop.

Figure 3. Theory of Operation Example (Repetitive Circuit Explanation).

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NOTE:

The Mating Jack Connector is a Mirror Image of the Connector Shown

Figure 4. Sample Connector Pin Numbers Illustration.

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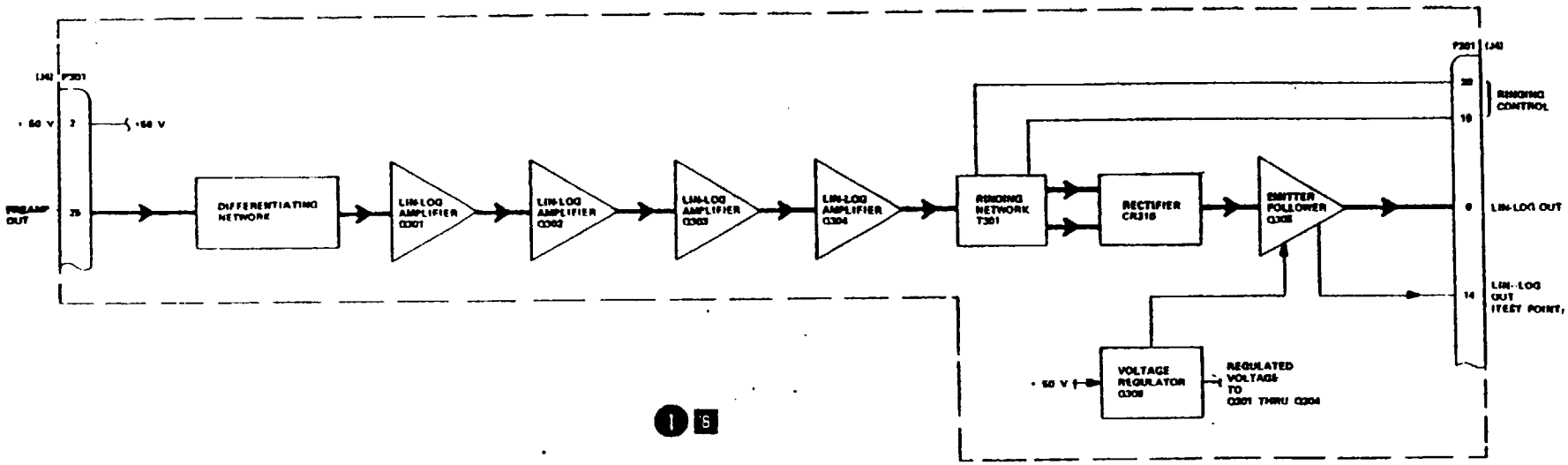


Figure 5. Module Schematic

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